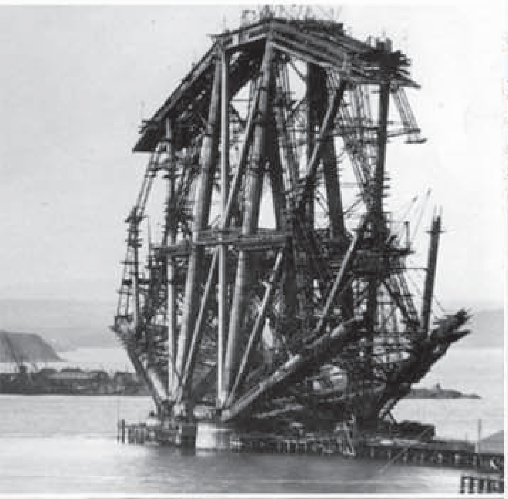
▽ **Forth Rail Bridge, Edinburgh**
Completed in 1890, this was the world's first steel cantilever bridge. It cost the lives of 57 of its 4,000 construction workers.



△ **British coal miners lead pit ponies**
Coal fueled the industrial revolution, and in Britain output increased almost sixfold in fifty years, from 11 million tons in 1800 to 65 million tons in 1854.

△ **Trade and transportation**
National and international transport networks were vital to the continued momentum of the Industrial Revolution. London's Albert Dock was opened in 1880 as the second of the Royal Group of Docks.

tied for life to work their landowner's land—which was abolished in France during the 1790s, in Germany from 1811 to 1848, and in Russia and Poland in the 1860s. In the US, immigrants moving to North America from Europe brought new skills and labor.

17 **The number of men employed by German steel manufacturer Krupp in 1826—this number rose to 122 in 1846 and 70,000 in 1910.**

The age of steam and steel

The second wave of industrialization was founded on new industrial enterprises: chemicals, engineering, and steel production, aided by the Bessemer process (see right).

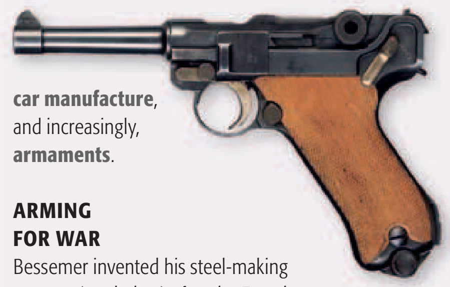
Railroads provided the momentum for continued industrialization. In addition to transporting raw materials and finished products, they also affected the economies of industrializing nations. Not only did they connect previously disparate economic regions, but financing railroads required new approaches to investment, such as a shift from private to joint-stock banking (see pp.276–77), which provided greater access to capital for industry in general. Similarly, the progression from sail to steamships had an impact on global trade. Foodstuffs and raw materials could be bought from the cheapest supplier, and the market for finished products increased. Improved communications, such as the invention of the telegraph and telephone (see pp.344–45), enabled businesses to respond relatively quickly to changes in the marketplace. They were also able to establish links with the farthest parts of the world which, thanks to the developments of the Industrial Revolution, had become intimately linked with their own.

AFTER

Until the advent of the steam locomotive, the fastest form of transportation on Earth was a galloping horse. Industrialization changed everything: the dominance of agriculture was over, cities now ruled, and the consumer society was born.

THE THIRD WAVE

After the first phase of the industrial revolution in Britain, and the second in Belgium, Germany, and the US, economic growth was slowed by a **worldwide depression**. Recovery was triggered by a **third wave of industrialization** from the 1890s onward in countries including Russia, Sweden, France, Italy, and Japan. Where the first wave had centered on textiles and iron production, and the second on heavy engineering and steel, the third wave saw the application of industrial processes to **chemical and electrical engineering**, **car manufacture**, and increasingly, **armaments**.



LUGER PISTOL

ARMING FOR WAR

Bessemer invented his steel-making process (see below) after the French complained that a **new artillery shell** he had invented for use in the Crimean War was **too powerful** for their cast-iron cannons. The advent of steel sparked an **arms race** that changed the face of warfare forever with the introduction of mass-produced guns, heavy artillery, and tanks.

OMINOUS SIGNS

German dominance in industrial production and weaponry toward the end of the 19th century led its increasingly nervous neighbors to accelerate industrialization. Russia, France, and Italy all **invested in arms manufacturing**, and Russia improved its railroad network specifically for transporting troops to defend its borders.

INVENTION

BESSEMER PROCESS

Steel-making was one of the key characteristics of the second phase of the Industrial Revolution. Previously, engineers had used cast iron (strong when compressed) or wrought iron (strong under tension). On October 17, 1855, building on previous investigations in this field, English metallurgist Henry Bessemer filed a patent for a means of producing mild steel by blasting cold air into molten iron in a "Bessemer Converter." This reduced the amount of carbon in the iron, making a stronger, more versatile product used for railroad lines, shipbuilding, and armaments.